

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1547

Rogers Lake 100 kV
June 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1547. The Interconnection Customer proposes a 143.9 MW battery energy storage facility interconnecting at the Rogers Lake 100 kV substation in the Ohio area. The request was received on August 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 418 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,898,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gilstrap 230/138kV TX	Rogers Lake 230-1 kV Autotransformer	126.6	129	21.1
Kepler – High Bridge 100kV line	Guayanes 2902 – Geauga Substation 100kV	120.6	150	6.0
Kepler – 88Th-Fruitridge 100kV	Guayane 2902 – Geaua 100kV Line	117.6	131	6.0
Rogers Lake 138/100 kV Autotransformer #1	Rogers Lake 138/100 kV Transformer T2	105.8	80	41.4
Rogers Lake 138/100kV Autotransformer Bank 2	Rogers Lake Substation 138/100kV Auto	105.8	80	41.4
Rogers Lake 138/100kV Transformer Bank T3	Rogers Lake 138/100kV XFMR Bank 1	105.8	80	41.4
88ThFruitridge – Hathaway 100kV	Pogressive – Wethersfield 100kV	101.9	137	13.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gilstrap 230/138kV TX	Replace transformer with higher-rated unit	\$11,551,000	44% (\$5,065,000)
Kepler – High Bridge 100kV L	Rebuild line with higher-capacity conductor	\$2,468,000	71% (\$1,759,000)
Kepler – 88Th-Fruitridge 100kV	Rebuild line with higher-capacity conductor	\$3,242,000	71% (\$2,311,000)
Rogers Lake 138-100 kV Transformer Unit 1	Replace transformer with higher-rated unit	\$4,216,000	100% (\$4,216,000)
Rogers Lake 138/100kV Autotransformer Bank 2	Replace transformer with higher-rated unit	\$5,116,000	100% (\$5,116,000)
Rogers Lake 138-100 kV XFMR #3	Replace transformer with higher-rated unit	\$6,021,000	100% (\$6,021,000)
88ThFruitridge – Hathaway 100kV	Reconductor line	\$410,000	100% (\$410,000)
	Total allocated to project		\$24,898,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.